5-2 Milestone Four: Enhancement Three: Databases

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This artifact is based on my original CS-210 Course Advising System, which was created as a simple C++ console application to load course data from a CSV file and allow users to view a full course list or query individual course information. At the time of its original development, the program was entirely file-based and temporary in nature, meaning all data was lost once the program stopped running. It served as an introduction to structured data handling, file parsing, and basic object-oriented programming principles.

I selected this artifact for my ePortfolio because it clearly demonstrates growth from a foundational programming assignment into a more advanced, database-driven application. It represents a strong opportunity to showcase improvements in data persistence, system design, and algorithmic problem solving. The original version was limited in scope, but it provided a solid baseline for demonstrating how a simple application can evolve into a more realistic software system used in academic or organizational environments.

The enhanced version of the artifact significantly improves both functionality and design by integrating an SQLite database to replace the original CSV-only storage approach. This change allows the system to persist course data beyond runtime and simulates a more realistic backend data storage solution. In addition to database integration, I expanded the system to support full CRUD functionality, allowing users to add, update, delete, and retrieve course records dynamically. This shift moves the program closer to a real-world advising system rather than a static data viewer.

Another major enhancement was the addition of a depth-first search (DFS) algorithm to validate prerequisite relationships and detect circular dependencies. This improvement demonstrates the application of algorithmic thinking and data structure traversal to ensure data integrity within the system. It also strengthens the reliability of the application by preventing invalid course dependency chains, which is an important consideration in academic advising systems.

This enhancement directly aligns with the course outcomes related to designing and evaluating computing solutions using algorithmic principles, as well as implementing software solutions using modern tools and practices. By combining structured data storage (SQLite), object-oriented design, and graph traversal algorithms, the project demonstrates an integrated approach to solving a meaningful data management problem.

Reflecting on the development process, the most significant learning outcome was understanding how application architecture changes when moving from file-based storage to a relational database system. I gained a better understanding of how data persistence impacts system design, especially in terms of structure, scalability, and maintainability. One of the main challenges was ensuring that database operations remained synchronized with the in-memory data structure while also maintaining program stability during updates and deletions. Another challenge was implementing the DFS logic in a way that remained efficient while handling edge cases such as missing or incomplete prerequisite data.

Overall, this enhancement strengthened both my technical and problem-solving skills. It also helped reinforce the importance of designing systems with scalability and real-world application in mind. The final result is a more robust, flexible, and professional-grade application that better reflects industry expectations for database-driven software systems.